

VIKASH MADDI

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EDUCATION

Indian Institute of Technology Madras, Chennai, India

July 2018 – May 2022

Bachelor of Technology in Mechanical Engineering

8.31/10

PROFESSIONAL EXPERIENCE

VMock, Full Stack Developer

August 2022 – Present

Scalable Backend Services

- Reduced API response latency from **400ms to 30ms** (13x) by analyzing query plans, adding targeted indexes, introducing Redis caching and refining service boundaries.
- Led a zero-downtime cross-datacenter migration of **90k+ accounts** across US and UK using dual-write, hash-based reconciliation and staged rollouts, achieving 99.9% data integrity.

Cloud & Container Operations

- Operated containerized services on Docker and **Kubernetes** across **AWS** (S3, SQS) with end-to-end CI/CD pipelines from repository to production.
- Ran event-driven workers on **AWS SQS** with explicit retry, timeout and idempotency semantics to ensure reliable background processing.

ML Platform Reliability & Observability

- Designed an agent tool-use framework over a structured registry, adding retries, timeouts and rate-limit handling with schema-validated guardrails.
- Implemented LLM observability using **Arize Phoenix**, adding span-level tracing across the agent lifecycle to attribute failures to model, prompt, tool or harness.
- Tracked token, cost and latency per session as platform metrics and added a prompt-caching layer that cut inference cost on long-running sessions.

Karomi Technologies, Deep Learning Intern

May 2021 – August 2021

- Trained Faster R-CNN, RetinaNet and EfficientDet models in PyTorch with semi-supervised learning, reaching **mAP 96%** for tables and **95%** for symbols.

PROJECTS

Context Management Proxy and MCP Server | *Python, MCP, Agent Infrastructure*

2026

- Built a transparent **HTTP proxy** and **MCP server** that compresses agent context without client changes, reducing token usage and cost on long-running tasks.
- Added relevance-based eviction and prefix-stable **prompt caching**, then validated cost reduction with an evaluation harness measuring cost per completed task.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, JavaScript, Bash, C++, PHP, Java

Agentic AI Stack: LangGraph, MCP (authoring and hosting servers), Tool-Use Frameworks, Agent Orchestration, Guardrails, OpenAI/Anthropic APIs, Context Management, Prompt Caching, Claude Code, Cursor

Backend & APIs: FastAPI, Flask, Laravel, REST APIs, Microservices, asyncio, Pydantic, API Versioning

Containers & Cloud: Kubernetes, Docker, AWS (S3, SQS, IAM basics), Cloudflare Workers, Linux, Deployment Automation, Rollback Strategy

Data & Retrieval: MySQL, PostgreSQL, Redis, Elasticsearch, Embeddings and Vector Retrieval, Chunking and Relevance Tuning, Event-Driven Pipelines, PyTorch

CI/CD & Tooling: GitHub Actions, Git-based Review, pytest, Self-Serve Internal Tooling, Policy-as-Code

AI Observability & Evaluation: Arize Phoenix, Span-Level Tracing, LLM-as-Judge Graders, Regression Suites, Token/Cost/Latency Tracking

SCHOLASTIC ACHIEVEMENTS

- Secured a global rank of 38 in the July Long Challenge on the CodeChef platform.
- Secured an all-India open category rank of 1308 in JEE-ADVANCED 2018 among 1.8 million.